OPERATION PHANTOM SWIPE

Technical-Legal Digital Forensics Investigation Report

Case ID	OPS-PS-2026-001
Investigation Type	Simulated Cross-Border ATM Skimming and Online Credit Card Fraud
Prepared For	Academic Digital Forensics Assignment
Evidence Status	SIMULATION ONLY

Scope: Cybercrime classification, simulated evidence acquisition, chain of custody, SHA-256 integrity verification, digital artefact analysis, controlled password recovery, legal mapping, and cross-border investigation considerations.

Important: All evidence and findings in this report are synthetic and were created for academic demonstration. No real financial, personal, confidential, or compromised information was used.

1. Executive Summary

Operation Phantom Swipe is a simulated digital-forensics investigation involving an alleged cross-border financial fraud ring. The scenario focuses on ATM skimming, credit-card cloning, and online credit-card fraud. Two simulated devices were considered: an ATM skimmer and a mobile-phone evidence set containing fraudulent-application data and communication records.

The investigation applies basic forensic principles including evidence preservation, chain-of-custody documentation, SHA-256 hashing, forensic searching, artefact extraction, and controlled dictionary-based password-recovery testing. The objective is to demonstrate a defensible forensic workflow rather than investigate a real criminal event.

2. Objectives

- Identify and classify the cybercrimes involved.
- Simulate seizure and acquisition of electronic evidence.
- Maintain a documented chain of custody.
- Calculate SHA-256 hashes for acquired evidence and use them for integrity verification.
- Search simulated devices for relevant digital artefacts, including strings, metadata, communications, location and transaction information.
- Perform a controlled dictionary-based password-recovery simulation.
- Examine legal, ethical, and cross-border investigative considerations.

3. Case Scenario

The simulated case involves a criminal group suspected of using ATM-skimming equipment to obtain payment-card information. The stolen information was allegedly used to support unauthorized payment activity. A second evidence set, representing a mobile phone, contains communication records, location information, device information and fraud-related application data.

The investigation therefore considers both physical electronic devices and the digital information stored or generated by them. Because the case is simulated, all artefacts are illustrative and do not establish facts about real persons or systems.

4. Cybercrime Classification

4.1 ATM Skimming

ATM skimming is classified as financial cybercrime and payment-card data theft. A skimming device can capture payment-card information during an ATM transaction. Relevant artefacts may include device configuration, logs, timestamps, device identifiers, location information and captured card references.

4.2 Credit-Card Cloning

Credit-card cloning involves unauthorized use or reproduction of stolen payment-card information. It can be classified as financial cybercrime, identity-related crime, payment-card fraud and data misuse. Relevant evidence may include transaction information, stored payment information and communication records.

4.3 Online Credit-Card Fraud

Online credit-card fraud involves unauthorized use of payment information through online systems. Possible evidence includes browser history, transaction records, IP addresses, login information, emails, messaging records and device identifiers.

5. Legal Analysis

The simulated activities may engage provisions of the Information Technology Act, 2000 depending on the facts established. The project identifies Sections 43, 66, 66C and 66D as potentially relevant. The assignment also requests historical IPC references: Sections 419, 420, 468, 471 and 120B. These references are presented for the assignment; exact applicability depends on the facts, jurisdiction and date of the alleged conduct.

For offences occurring after 1 July 2024, the current criminal-law framework must also be considered, including the Bharatiya Nyaya Sanhita, 2023. The report does not treat any single provision as automatically applicable.

5.1 International Legal Considerations

Because the scenario is cross-border, cooperation may be required where evidence involves foreign-hosted servers, overseas cloud storage, international payment systems, foreign IP addresses, cross-border communications or international financial transactions. The Budapest Convention on Cybercrime provides an international framework concerning cybercrime and electronic evidence, while formal legal procedures may be required to obtain evidence held in another jurisdiction.

6. Digital Forensics Evidence

Two simulated devices were examined. Device 01 represents an ATM skimmer and contains configuration and device information. Device 02 represents a mobile phone and contains messages, GPS information, device information and simulated transaction information. The evidence is maintained in the repository's evidence directory.

7. Evidence Acquisition

The investigation uses a simulated acquisition process. Evidence files are treated as forensic copies rather than original live evidence. A write-protection principle is considered during acquisition. In a real investigation, appropriate forensic write blockers should be used when handling original storage media to reduce the risk of accidental modification.

The acquisition process is documented using a chain-of-custody record.

8. Chain of Custody

The chain-of-custody record documents case identification, evidence identification and description, collection information, transfers, storage, examiner information and integrity verification. Maintaining chain of custody helps demonstrate that evidence was handled systematically and that its integrity was monitored throughout the investigation.

Repository file: acquisition/chain_of_custody.md

9. Hashing and Evidence Integrity

SHA-256 was used to calculate cryptographic hashes for the simulated evidence. The hashing script is scripts/hash_files.py and the resulting values are stored in acquisition/hashes_sha256.txt.

A cryptographic hash provides a fixed-length representation of digital data. If file contents change, the resulting SHA-256 hash will normally change. Comparing recorded and later-calculated hashes therefore provides a practical method for detecting modification of forensic copies.

10. Digital Artefact Analysis

The simulated evidence was examined using string searching, file inspection, metadata review, communication-record examination, location information and transaction information. Extracted artefacts are documented in analysis/extracted_artifacts.csv, with supporting discussion in analysis/evidence_analysis.md.

11. Sample Digital Artefacts

- Device information and identifiers
- Configuration information and logs
- Communication records
- GPS coordinates and timestamps
- Simulated transaction information
- Other event-related strings and metadata

Taken together, these artefacts can help investigators establish relationships between devices, users, locations and events. In an actual case, conclusions would require validation, corroboration and appropriate attribution procedures.

12. Cryptography and Password Recovery

A controlled password-recovery experiment was performed using synthetic evidence. The simulated password phantom2026 was converted into a SHA-256 hash. A dictionary containing possible passwords was then tested by calculating each candidate's SHA-256 hash and comparing it with the target hash. The matching candidate was successfully identified, demonstrating the basic principle of a dictionary attack.

13. Ethical Considerations

Password recovery must only be performed with appropriate legal authority and within the authorized investigative scope. Investigators should not attempt to access real devices, accounts or data without authorization. In this assignment, recovery was performed only against synthetic data created for educational purposes.

14. Password Security Observation

The simulation demonstrates the risk of predictable passwords. Long and unique passwords, password managers and multi-factor authentication reduce exposure to simple dictionary-based guessing. Predictable words, names and number patterns should be avoided.

15. Cross-Border Investigation Challenges

- Jurisdiction: countries may apply different laws and investigative procedures.
- Evidence location: electronic evidence may be stored in another country.
- Evidence preservation: digital information can be deleted or modified rapidly.
- Legal requests: formal international procedures may be necessary to obtain evidence.
- Privacy: cross-border collection must consider applicable privacy and data-protection requirements.
- Time zones: original timestamps should be preserved and any timezone conversion documented.

16. Evidence-Handling Challenges

- Preventing accidental modification of original evidence.
- Maintaining evidence integrity through hashing and controlled access.
- Correctly documenting every evidence transfer and examination.
- Managing large quantities of digital data and identifying relevant artefacts.
- Maintaining reliable timestamps and documenting timezone conversions.
- Protecting sensitive evidence from unauthorized access.

17. Recommended Law-Enforcement SOPs

- Standardized acquisition: use documented procedures for seizure and forensic imaging.
- Write protection: use appropriate hardware or software write-protection methods when handling original storage media.
- Cryptographic verification: calculate SHA-256 or another approved cryptographic hash during acquisition and verification.
- Detailed chain of custody: record every evidence transfer, examination and storage event.
- Automated evidence logging: standardize recording of file paths, timestamps and hash values.
- Cross-border coordination: establish rapid communication procedures with international counterparts and relevant service providers.
- Evidence preservation: request preservation of relevant electronic information as early as legally permitted.
- Examiner training: maintain training in mobile, cloud, network and financial forensics.

18. Results

The simulated investigation successfully demonstrated cybercrime classification, evidence acquisition, chain-of-custody documentation, SHA-256 hashing, digital artefact extraction, evidence analysis, dictionary-based password recovery, legal mapping and cross-border investigation considerations. The simulated password was recovered from the dictionary, illustrating the effectiveness of dictionary testing against a weak predictable password.

19. Limitations

This project is an academic simulation. It contains no real credit-card information, banking credentials, victim information, criminal communications, stolen data or compromised devices. Accordingly, the findings demonstrate forensic methodology rather than conclusions about a real criminal investigation.

20. Conclusion

Operation Phantom Swipe demonstrates foundational principles of digital forensics through a simulated cross-border financial-crime investigation. The project shows how investigators can classify cybercrime, preserve electronic evidence, calculate cryptographic hashes, analyze digital artefacts and document findings. It also demonstrates the importance of legal authorization, chain of custody and evidence integrity.

The cross-border nature of modern cybercrime highlights the need for international cooperation and efficient electronic-evidence preservation mechanisms. Overall, the investigation provides a practical introduction to evidence handling, analysis, cryptography and legal considerations.

References

- Information Technology Act, 2000 (India), as amended.
- Indian Penal Code, 1860 / Bharatiya Nyaya Sanhita, 2023 (India).
- Council of Europe, Convention on Cybercrime (Budapest Convention), ETS No. 185, 2001.
- Project files: acquisition/chain_of_custody.md; acquisition/hashes_sha256.txt; analysis/evidence_analysis.md; analysis/extracted_artifacts.csv; cryptography/cracking_report.md.